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Liquid Universe Continuation Strategies for Crypto Perpetual Futures

Executive conclusion and strategy map

The strongest starting point is not a single "breakout pattern," but a hierarchy. The best-supported and most depth-feasible family is **time-series momentum on liquid perpetuals with volatility management and crowding vetoes**. Next come **prior-high proximity and liquid-leader breakout continuation** in large, well-traded names. After that, **post-breakout retest continuation** is reasonable as a practitioner extension of the same effect, but its direct crypto-specific literature is thinner. **Pure cross-sectional winner-minus-loser momentum** is less reliable once implementation frictions matter, and **sector-leader continuation** should be treated as an optional overlay only if you can freeze a credible sector taxonomy through time. [1]

That ranking follows the evidence. Bitcoin, Ethereum, and other major coins show time-series momentum over daily to weekly horizons; large and liquid coins show continuation more often than reversal; a direct crypto paper now finds a positive nearness-to-52-week-high effect; and a recent technical-factor paper shows that aggregated trend signals remain predictive even in the largest and most liquid coins and survive transaction-cost tests. By contrast, more recent practitioner-academic work argues that once daily path effects, transaction costs, and liquidations are treated more realistically, long-only time-series momentum remains, while cross-sectional momentum becomes much weaker. [2]

For a small research team with limited historical depth, this is good news. The family you want to study is the family least dependent on queue modeling: **liquid-name, slower-horizon, continuation trades whose signal is still visible in OHLCV, mark/index, funding, open interest, and venue fee schedules**. That is much more defensible than small-cap event pumps, short-horizon mean reversion, or order-book imbalance strategies, where the edge and the cost model both live inside the tape. The evidence that momentum concentrates in larger, more liquid coins and that small, illiquid names are where reversal and implementation problems dominate is especially important here. [3]

The practical strategy map is therefore:

- **Tier one**: volatility-managed time-series momentum on liquid perpetuals, with BTC/ETH parent filters and funding/OI crowding vetoes. [4]
- **Tier two**: prior-high proximity and liquid-leader breakout continuation, restricted to large, liquid names. [5]
- **Tier three**: breakout-retest/reclaim continuation and impulse-plus-base breakout, treated as implementation-friendly decompositions of the same trend effect rather than as independent anomalies. The support here is more indirect, coming from crypto trend-factor and path-continuity evidence plus standard momentum literature. [6]
- **Tier four**: sector-leader continuation if, and only if, you have a frozen, versioned sector dataset. CoinGecko documents more than 600 categories and notes that some cryptocurrencies can overlap

across categories, which is useful for idea generation but dangerous for historical research unless you snapshot the taxonomy. [7]

What the evidence says about crypto momentum

There is now a fairly broad base for **time-series momentum** in crypto. Liu, Tsyvinski, and Wu document statistically significant time-series momentum in Bitcoin at daily and weekly horizons, with weekly formation signals carrying through one to four weeks ahead; their out-of-sample checks also retain the effect. The same paper finds related, though less uniform, evidence for Ethereum and Ripple. This is the cleanest academic foundation for “trend-follow liquid names.” [8]

More recent work expands the same direction. A 2026 study titled *Momentum in the Cryptocurrency Market* argues that prior work overstated implementable profits and that **long-only time-series momentum** is the only crypto momentum approach that remains robust once daily fluctuations, transaction costs, and liquidations are incorporated; it reports that cross-sectional momentum and time-series long-short momentum become largely unprofitable under those more realistic assumptions. A newer comparative study also reports that time-series momentum outperforms cross-sectional momentum on a risk-adjusted basis in major cryptocurrencies. The exact realized magnitudes should be treated carefully because these are recent papers with different samples and assumptions, but the directional message is consistent: if you must choose one family first, choose long-biased time-series continuation, not market-neutral relative-strength spreads. [9]

The literature on **cross-sectional momentum** is mixed rather than absent. Practitioner work by Drogen, Hoffstein, and Otte finds a short-term anomaly in which assets that outperform over 30 days tend to continue to outperform over the next 7 days, and that long-only momentum strategies beat Bitcoin as a benchmark in their sample. A separate academic line goes further: the CTREND paper uses 28 technical signals aggregated across prices, volume, and volatility, and finds a cross-sectional trend factor that predicts returns, is robust across subperiods and research designs, works even in the largest and most liquid coins, and remains resilient to transaction costs up to holding periods of roughly four weeks. That does not rescue simple winner-minus-loser momentum automatically, but it does support **top-relative-strength within a liquid universe** as a plausible research branch, especially if you use long-only or long-biased implementations instead of symmetric long-short portfolios. [10]

For **prior-high proximity**, the case is now much stronger than it was even a year ago. Jia and coauthors study the nearness to the 52-week high in cryptocurrencies and find a significant positive association with subsequent cross-sectional returns. Their value-weighted long-short cANCHOR portfolio earns about 130 basis points per week, and the effect survives controls for other predictors and alternative specifications. That is exactly the academic bridge you need from the equity 52-week-high literature to a crypto “prior-high / yearly-high analog” strategy. [11]

The **size/liquidity interaction** is one of the most useful findings for your use case. Fičura shows that weekly reversal is concentrated in small and illiquid coins, while large and liquid coins exhibit weekly momentum, and that the short-term reversal effect is heavily linked to very low dollar volumes. Begušić and Kostanjčar similarly find a strong momentum effect among the most liquid cryptocurrencies. This is not just an academic nuance. It means a liquid-universe continuation strategy is not merely easier to execute; it is also better aligned with where the anomaly has actually been observed. [12]

The **technical-trend literature** also matters because your data are bar-centric. CTREND explicitly argues that the information in past prices and volume is richer than simple return lookbacks and shows that aggregate technical signals can explain crypto cross sections better than standard benchmark models. Separately, a new paper on crypto price path continuity applies the "frog-in-the-pan" idea to cryptocurrencies and finds that smoother, more continuous prior paths are more likely to continue, while short-term past returns on average can reverse unless filtered by path continuity. That is the academic reason to include a **smooth path filter** or **impulse-plus-base structure** rather than treating every sharp spike as equal. [6]

On **intraday versus multi-day horizons**, the warning is straightforward. Wen and coauthors document both intraday momentum and intraday reversal in Bitcoin and other actively traded cryptocurrencies, and show that the pattern shifts with jumps, FOMC releases, and liquidity states. Caporale and Plastun find intraday and next-day continuation after one-day abnormal returns in BTC, ETH, and LTC, but with some asymmetry and occasional contrarian cases. The implication is not that intraday continuation does not exist. It does. The implication is that sub-hour and intraday strategies are more entangled with microstructure and event timing, while multi-hour to multi-day continuation is the cleaner choice for a team that does not yet have full historical depth and trade data. [13]

Failure modes and regime activation

Momentum in crypto fails for familiar reasons, but often in sharper form. The general cross-asset literature shows that momentum crashes tend to arrive in "panic" states, after declines and during high volatility, and often coincide with violent rebounds. A new crypto paper finds the same basic issue: cryptocurrency momentum is subject to severe crashes, tail variance can be effectively unbounded in practice, and volatility-scaling materially improves the strategy. That is the core reason your first production-worthy contract should be **vol-managed** rather than static notional. [14]

In crypto perps, **crowding** has venue-native observables rather than just latent sentiment. Funding is the transfer mechanism that keeps perp prices near spot. Bybit states that the funding rate is updated every minute and settled according to each symbol's interval; Kraken's EEA perpetual specifications describe the funding rate as a time-weighted average premium standardized per hour with continuous accrual and hourly settlement; Coinbase International defines funding from a one-hour TWAP premium with smoothing over prior periods and charges it hourly. A 2026 paper on perpetuals frames funding not as a passive transfer, but as an algorithmic feedback rule. That means extreme positive funding is not just "bullish mood"; it is a direct measurement of how much the market is paying to maintain long exposure. Inference: continuation entries become materially worse when you chase breakouts with already-extreme funding. [15]

Open interest should be treated the same way: as a state variable, but normalized carefully. Bybit's open-interest display changed on June 11, 2026 from double-sided to single-sided, which mechanically cuts displayed OI by about half without changing actual market activity. Bybit's API also shows that historical OI fields now include both openInterest and singleOpenInterest, and for linear USDT contracts the unit is the base asset rather than USD. If you compare raw OI across time or venues without normalizing units and methodology, you will create fake signals. For research, use **changes, z-scores, and OI-to-volume or OI-to-market-cap style ratios**, not absolute contemporaneous levels. [16]

Breadth and liquidity concentration are the next major failure mode. CF Benchmarks noted in mid-2025 that large-cap crypto performance had become highly concentrated in Bitcoin and Ether. Kaiko's Q1 2025 report shows Bitcoin depth staying resilient while ETH liquidity fell 27% over the quarter and top-50 altcoin depth fell 30%, with liquidity concentrating further in large-cap alts. Kaiko's earlier exchange-level liquidity report also found that order-book depth and volume are heavily concentrated across a small set of venues. For your strategy family this means: continuation works best when trend is expanding across liquid names, not when one or two majors carry the tape while the rest of the universe gets thinner. [17]

Listing mania is a separate problem and should be treated as an exclusion rule, not a regime filter. Exchange-listing event studies find large positive abnormal returns around listings, but those moves are event-driven, exchange-specific, and prone to informed flow and severe post-event distortions. That is the wrong ecology for a fair bar-data continuation backtest. A reconstructed historical universe should therefore include lifecycle metadata and a cooling-off period after listing or major contract launch. [18]

The simplest **regime activation map** is the one most consistent with both the literature and the available data:

In the best regime, BTC and preferably ETH are themselves above medium-term trend; liquid-universe breadth is expanding; altcoin funding is positive but not extreme; OI is rising but not in a euphoric one-way burst; and liquidity is broad rather than collapsing into a handful of majors. This is when you want leader breakouts, prior-high setups, and retest continuation fully active. [19]

In the acceptable regime, BTC is trending, ETH is neutral, breadth is mixed, funding is modest, and realized volatility is elevated but not disorderly. Here I would keep **time-series momentum with vol targeting** active, keep **prior-high continuation** active in universe A, and reduce or disable more pattern-sensitive breakout/retest contracts in universe B. [20]

In the bad regime, BTC/ETH trend is rolling over, breadth is narrow, altcoin liquidity is falling, funding is already crowded, and OI is expanding faster than price quality. This is where momentum crashes, squeeze risk, and fake breakouts cluster. In that state, new long continuation should be disabled or heavily de-rated unless the strategy is explicitly designed as a low-gross, volatility-scaled parent-trend model. [21]

Recommended first strategy contracts

The first five contracts should be kept deliberately close to the evidence. They should also share a common implementation stack so you can compare them fairly.

Volatility-managed liquid-universe time-series momentum should be the first contract. Use a liquid universe only, one or more medium-horizon lookbacks, and scale target exposure inversely to recent realized volatility. This is the best-backed strategy in the literature and the least dependent on precise microstructure modeling. The rationale comes from the direct crypto TSMOM evidence and from the finding that crypto momentum crashes can be materially improved by volatility management. [22]

A sensible initial range is: signal lookbacks of 20, 40, 60, and 90 trading days; volatility estimates over 20 or 30 days; rebalance daily or every 8 hours; and parent filters requiring BTC or BTC+ETH to be above their own 50-day or 100-day trend. Exits should be compared across fixed holds, 10/20 EMA trail, and time stops.

Those values are not “magic”; they span the horizons where the literature finds continuation while staying slow enough for conservative taker modeling.

Prior-high proximity momentum should be the second contract. The academic anchor is the Nearness52 result. The trade definition should be long-only: buy liquid names that are within a narrow band under their rolling 252-day high, but only when the parent market filter is on and funding/OI are not in the most euphoric tail. A practical parameter grid is 0% to 5% below the 252-day high, with optional shorter analogs such as 90-day and 180-day highs for newer contracts. The exit comparison should again be fixed hold versus EMA trail versus time stop. [23]

Liquid leader breakout long should be the third contract. This is the most intuitive expression of the large-liquid continuation evidence. Define a “leader” as a name in the top relative-strength bucket over a trailing 20 to 60 day window, subject to liquidity and age filters. Trigger on a close above the highest high of the prior 20, 55, or 90 bars, but use mark-price confirmation where available to reduce wick contamination. Because Bybit provides both mark-price and index-price klines, you can test last-trade breakouts against mark-confirmed breakouts directly. [24]

For this contract, add two structure filters. First, a **smooth path filter**: require the advance into the breakout to be gradual rather than a one-candle vertical move. The academic backing comes from crypto price-path continuity and the older frog-in-the-pan literature. Second, a **volatility contraction / base** filter: require realized range or ATR percentile to compress into the breakout. The literature here is more indirect, but it is consistent with the same underreaction logic and with the CTREND finding that aggregated technical signals outperform single-indicator momentum. [25]

Post-breakout retest continuation should be the fourth contract. This one is not a separate documented anomaly so much as a way to make the breakout trade more robust under conservative execution assumptions. Define an initial breakout, allow a pullback of some fraction of ATR or a reclaim of the breakout pivot after one to ten bars, then enter only if price reasserts. The reason to like this contract is practical: it reduces buying the maximum extension and tends to lower effective slippage per unit of ex ante edge. The evidence base is the combination of breakout-supporting momentum research, path-continuity evidence, and the fact that one-day abnormal-return continuation can persist into the next day. [26]

Sector-leader continuation should be the fifth contract, but only as conditional research. If you use CoinGecko or another provider for sector labels, you must version the taxonomy by date and decide ex ante how to handle overlaps. CoinGecko documents that categories exist at large scale and that assets may overlap across categories. Grayscale also publishes a six-sector framing, which is useful conceptually, but not a substitute for frozen daily membership histories. The contract itself should be simple: sector breadth positive, sector leader in top relative-strength bucket, BTC/ETH parent filter on, crowding veto off. Without that data governance, skip it. [27]

The common **funding/OI crowding veto** across these contracts should be percentile-based, not absolute. A strong default is to veto new longs if funding is above the rolling 90th to 95th percentile for that symbol and venue, or if funding is elevated and OI has expanded by an unusually large amount over the past one to five days while breadth is weak. This is partly an inference from funding/OI mechanics, not a fully settled theorem, but it is the correct way to keep your continuation family from becoming a “buy-the-squeeze-top” family. [28]

The common **failure / rejection criteria** should be strict. Reject a contract if edge disappears under taker/taker plus realistic slippage, if performance is concentrated in a few coins or one cycle, if the strategy only works in names that fail your liquid-universe thresholds, or if out-of-sample performance depends on a single funding/OI cutoff. Those are not literature facts; they are the minimum conditions for a contract that can survive live trading.

Data contract and execution model

A research-ready data contract is now clearly feasible on Bybit and partly portable. Bybit's public APIs provide historical candles, mark-price klines, index-price klines, historical funding, open interest, and instrument metadata including launchTime and deliveryTime. Kraken's derivatives stack provides market candles, historical funding rates, instruments with openingDate, and a market-analytics endpoint that includes open interest, trade volume, liquidation volume, spreads, liquidity, slippage, and more. Coinbase International provides instrument candles, historical funding rates, instrument details, and daily trading volumes; its trading rules define mark price, index price, and funding. That is enough to build a serious **bar + mark/index + funding + OI** research environment across venues, even if venue-specific normalization still matters. [29]

The minimum schema should include: OHLCV on at least 1m, 5m, 1h, 4h, and 1d bars; mark-price and index-price bars on the same grid; funding history with raw interval metadata; open interest with raw units and venue methodology flags; fee schedules by venue and effective date; instrument metadata with launch date, delist date, tick size, lot size, and leverage parameters; and a reconstructed symbol history so the universe can be rebuilt point-in-time rather than from today's survivors. Bybit's launchTime/deliveryTime and Kraken's openingDate are especially useful for that last requirement. [30]

For venue portability, the main issue is not candles. It is **mechanics normalization**. Bybit uses a dual-price system for perpetuals in which mark price reflects a global spot index plus a funding basis and is used for liquidation, and it documents that mark/index handling can change under abnormal conditions. Kraken uses impact-mid based premium logic with continuous accrual and hourly settlement. Coinbase International uses a median-based mark price and a smoothed premium TWAP updated hourly. Therefore, never compare raw funding rates, raw OI levels, or raw mark/last discrepancies across venues without transforming them into standardized, horizon-consistent features. [31]

Your **liquidity filters** should be rolling and point-in-time. The evidence points to two usable universe tiers. Universe A should be the highest-liquidity names that pass age, notional-volume, and OI thresholds. Universe B can include the next layer of liquid alts, but must carry harsher slippage and stricter breadth/funding vetoes. This is consistent with the academic result that continuation belongs to large/liquid names and with Kaiko's evidence that altcoin liquidity can disappear quickly even while BTC remains tradable. [32]

A conservative **execution model** for backtests should start with **taker on entry and taker on exit for every trade**, even if you intend to work passive orders later. Then add a stress layer in basis points that is tiered by universe and bar frequency. A good first pass is to test at least three slippage buckets per contract, with universe B carrying materially higher stress than universe A. Participation should be capped against recent 1m or 5m notional volume, with the stricter of the two caps applied. This is a recommendation rather than a published fact, but it is justified by the observed concentration of liquidity, by the instability of altcoin depth, and by the fact that your chosen family is meant to be slower and liquid enough that a taker/taker stress test is meaningfully conservative rather than absurd. [33]

Why is this family **less depth-sensitive** than small-cap event/reversal trading? Because the literature says the anomaly itself is stronger in large and liquid names, while reversal in small/illiquid names is partly a low-volume artifact and harder to realize in practice. Also, intraday crypto predictability changes as a function of liquidity and jumps, which is exactly what you want to avoid in a first-generation backtest. That does not mean depth does not matter. It means the signal-to-cost ratio of these continuation contracts is more likely to survive conservative bar-based assumptions. [34]

Before live trading, you still need **trade and order-book data** for three things: spread and queue-position realism, latency-sensitive stop behavior, and trade-through/slippage modeling in stressed conditions. Bybit publishes downloadable historical order-book data; Kraken's market analytics expose spreads, liquidity, slippage, and orderbook analytics; Coinbase's FIX market-data feeds expose orders and trades. That means you can backtest the family now, but you should not sign off on live deployment until you have at least partial L2 or trade-level replay for the exact symbols and hours you plan to trade. [35]

Validation protocol and implementation prompt

The validation stack should be harsher than the strategy logic. Use a reconstructed point-in-time universe, evaluate contracts separately in universe A and universe B, and log every tested parameter combination so multiple testing can be corrected explicitly. White's Reality Check and Hansen's SPA remain the right mental model for strategy data-snooping control, while Harvey-Liu and the Deflated Sharpe Ratio are useful for haircutting performance claims under multiple testing and non-normal returns. [36]

For model validation, use **rolling walk-forward** and **purged / embargoed cross-validation**, and if you can operationalize it cleanly, use **combinatorial purged cross-validation** to generate multiple independent backtest paths rather than a single flattering one. The reason is simple: financial labels overlap in time, and standard IID cross-validation will leak information. Recent work on backtest overfitting continues to endorse CPCV precisely for this setting. [37]

The event-study style contracts need **event counts and matched nulls**. Do not accept a breakout or retest strategy because its top decile of cases looks beautiful. Require enough events across symbols and market states, and compare each triggered event to matched non-events from the same symbol, volatility regime, and parent-trend regime. This part is a recommended protocol rather than a specific finding from a paper, but it is the only fair way to decide whether "breakout + smooth path + no crowding" beats simply buying strong names in strong markets.

The easiest answer to "what can be tested now" is:

Test now with bars + mark/index + funding + OI
Time-series momentum with vol management, prior-high proximity momentum, liquid-leader breakout long, breakout with smooth-path filter, and retest/reclaim continuation with conservative taker/taker costs. These rely mainly on bar construction, parent market filters, and crowding controls that are explicitly available in the current data stack. [38]

Delay until depth or trade data are available
Very short-horizon intraday continuation, spread-capture and passive-maker logic, any queue-priority dependent fill model, order-book imbalance entries, liquidation-chase variants, and aggressive small-cap/

event-driven continuation around listings or squeezes. Those are exactly the areas where the signal and the cost model collapse into the same microstructure problem. [39]

The best exact prompt for a backtesting AI agent is below. It is written to implement the first test, which should be the volatility-managed liquid-universe time-series momentum contract because that is the most strongly supported and the least likely to produce a false positive from missing depth.

```
You are implementing a research-grade backtest for crypto perpetual futures
continuation strategies.

Objective:
Build the first contract only:
VOL-MANAGED LIQUID-UNIVERSE TIME-SERIES MOMENTUM LONG

Research constraints:
- Use only data that can be fairly supported by bars and venue reference data.
- Do NOT use order-book imbalance, queue modeling, trade prints, or passive fill
assumptions.
- Assume taker entry and taker exit on all trades.
- Add configurable slippage stress in basis points on top of venue fees.
- Use point-in-time universe reconstruction only.

Input data expected:
1. OHLCV bars for perpetual contracts at 1h, 4h, and 1d.
2. Mark-price bars aligned to the same timestamps.
3. Index-price bars aligned to the same timestamps.
4. Funding-rate history with raw funding interval metadata.
5. Open-interest history with raw units and venue methodology metadata.
6. Instrument metadata with launchTime/openingDate, delist/deliveryTime, tick
size, lot size, quote coin, settle coin.
7. Fee schedule table by venue and effective date.

Target venue:
- Primary: Bybit linear USDT perpetuals.
- Code should allow later portability to Kraken and Coinbase International by
abstracting venue mechanics.

Point-in-time universe rules:
- Reconstruct eligible symbols by date using launch/delist metadata.
- Exclude symbols with age < 180 calendar days.
- Exclude symbols without complete bar/mark/funding/OI coverage for the
formation window.
- Define Universe A and Universe B by rolling liquidity:
  - Universe A: top liquid tier by rolling median daily notional volume and
rolling median OI.
  - Universe B: next liquid tier passing minimum thresholds.
- Keep thresholds configurable.
```

```
- Persist daily eligible universe membership to avoid survivorship bias.

Signal definition:
- Compute time-series momentum on daily bars.
- Test lookbacks: 20, 40, 60, 90 trading days.
- Long signal = 1 if close_t / close_{t-L} - 1 > 0, else 0.
- Optional parent filter modes:
  1. none
  2. BTC filter: BTC close above its 50d EMA
  3. BTC+ETH filter: both BTC and ETH above their 50d EMA
- Optional crowding veto:
  - veto if funding is above rolling percentile threshold for symbol
  - veto if OI change over lookback window exceeds rolling percentile threshold
AND funding is elevated
- All crowding thresholds must be percentile-based, not hard-coded raw values.

Risk management:
- Volatility target using inverse realized volatility scaling.
- Realized vol windows: 20d and 30d.
- Position weight proportional to target_vol / realized_vol, clipped to max
leverage or max notional per symbol.
- No shorting in this first contract.
- Cap portfolio gross exposure and per-symbol weight.
- Use configurable rebalance frequencies: daily close and every 8h if aligned
data exist.

Execution model:
- Enter at next bar open or next bar mark, selectable by parameter.
- Exit at next bar open or next bar mark, selectable by parameter.
- Apply venue fee schedule on both entry and exit.
- Apply slippage stress grid separately for Universe A and Universe B.
- Enforce participation caps using trailing 1m or 5m notional volume if
available; otherwise flag missing.
- If intrabar stop logic would require unsupported microstructure assumptions,
do not use it in this first contract.

Exit variants to compare:
1. fixed hold: 5d, 10d, 20d
2. trailing exit: close below 10 EMA or 20 EMA
3. time stop: max hold 20d

Backtest design:
- Run separately for Universe A and Universe B.
- Run for full sample, bull/risk-on regimes, and weak-breadth regimes.
- Compute rolling walk-forward results.
- Implement purging and embargo for validation splits.
- If feasible, implement CPCV and report distribution of out-of-sample Sharpe
and drawdown.
```

```
Performance outputs:
- Net returns after fees and slippage.
- CAGR, Sharpe, Sortino, Calmar, max drawdown.
- Hit rate, average gain/loss, payoff ratio.
- Turnover, holding time, event count.
- Contribution by symbol and by year.
- Sensitivity tables across lookback, vol window, parent filter, crowding veto,
exit, and slippage stress.
- Stability diagnostics:
  - performance concentration by year
  - performance concentration by top 3 symbols
  - degradation from no-cost to fee-only to fee+slippage

Failure criteria:
- Reject configuration if event count is too low.
- Reject if net edge disappears under conservative stress.
- Reject if PnL is dominated by a single symbol or single year.
- Reject if signal only works in Universe B and fails in Universe A.
- Reject if out-of-sample performance is materially worse than in-sample after
purged validation.

Implementation requirements:
- Use clean modular Python.
- Separate modules for data loading, universe construction, feature engineering,
signal generation, portfolio construction, execution assumptions, metrics,
validation, and reporting.
- Every transformation must be timestamp-safe and point-in-time.
- Include unit tests for:
  - universe reconstruction
  - funding and OI normalization
  - lookback alignment
  - fee schedule application
  - no-lookahead guarantees

Final deliverables:
1. reproducible backtest code
2. YAML or JSON config for all parameter grids
3. CSV or parquet output of trades and daily portfolio NAV
4. markdown report summarizing:
   - best configurations
   - out-of-sample robustness
   - sensitivity to slippage
   - whether this contract is production-candidate, research-candidate, or
rejected
```

Open questions and limitations

The evidence base is strongest for time-series momentum, technical trend, and 52-week-high style continuation. It is weaker and more indirect for **post-breakout retest** specifically, and much weaker for **sector-leader continuation** unless you can lock a sector taxonomy historically. That does not make those ideas bad. It means they should be treated as structure-preserving implementations of better-documented momentum effects rather than as standalone anomalies with equal evidentiary weight. [40]

The other unresolved tension is between what is **research-feasible now** and what is **live-ready**. The available bar, mark/index, funding, OI, and instrument metadata are enough to rank continuation contracts fairly. They are not enough to sign off on exact execution quality, stop behavior, or maker-versus-taker optimization. The missing piece is not signal discovery anymore. It is the microstructure audit that begins where the bars end. [41]

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